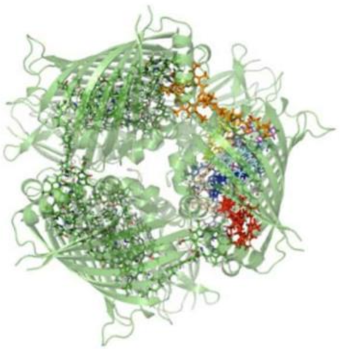
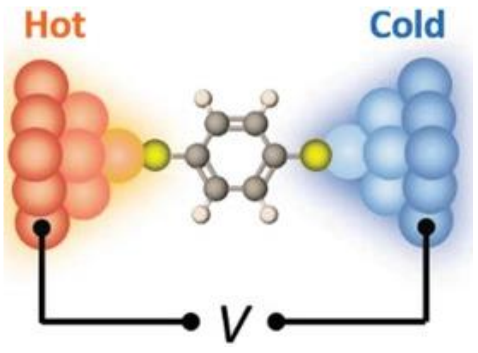
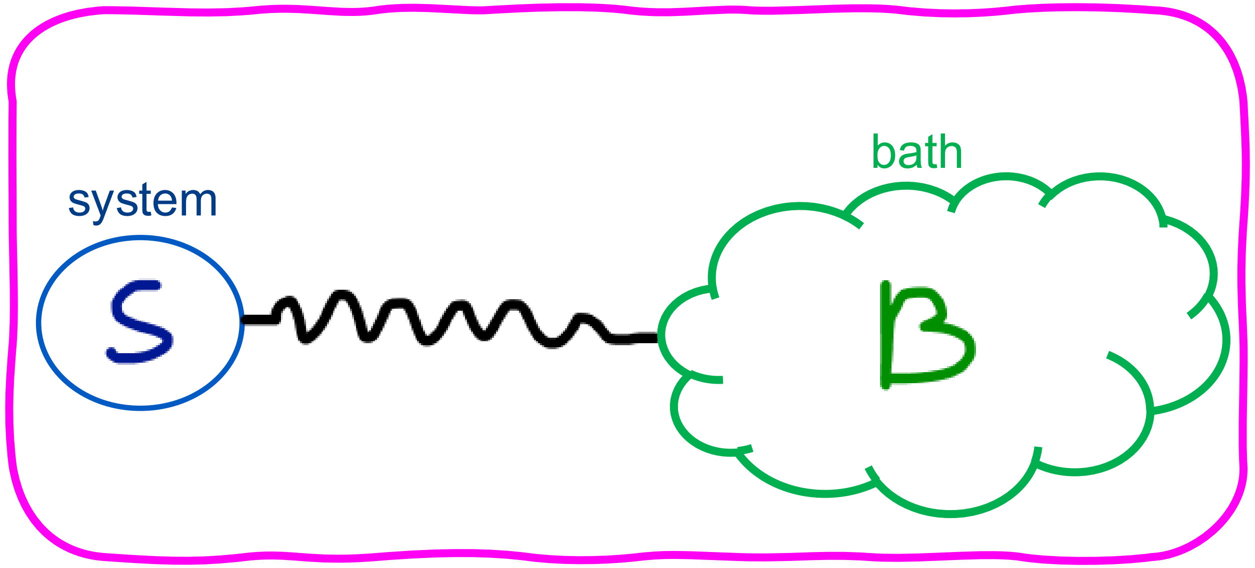
Open quantum systems: Intro

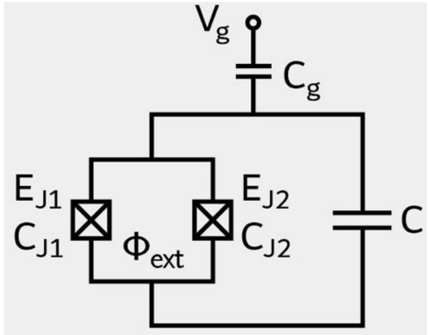


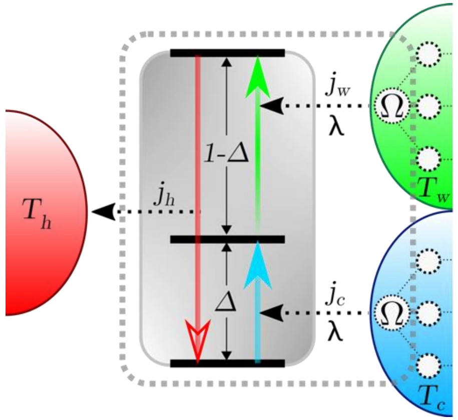
Dvira Segal
Departments of Chemistry and Physics
Director: CQIQC, QAT
University of Toronto

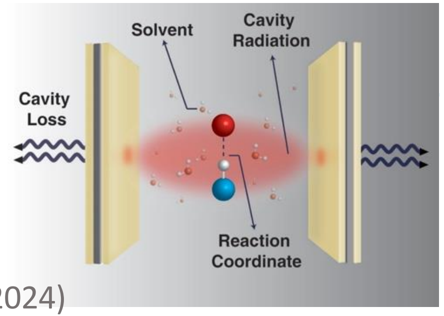


Advanced Functional Materials
Thermal and Thermoelectric Properties of Molecular Junctions
K Wang, E Meyhofer, P Reddy
https://doi.org/10.1002/adfm.201904534

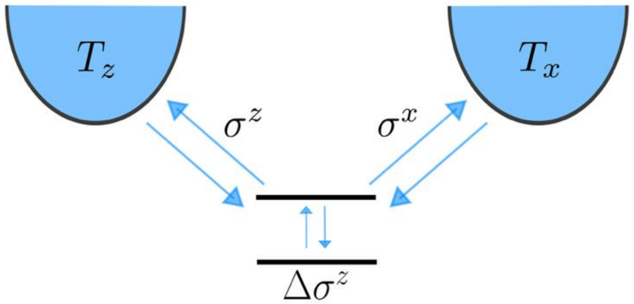


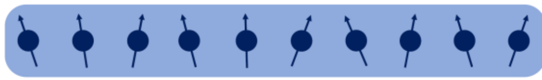






W Ying, P Huo
Communications Materials 5, 110 (2024)

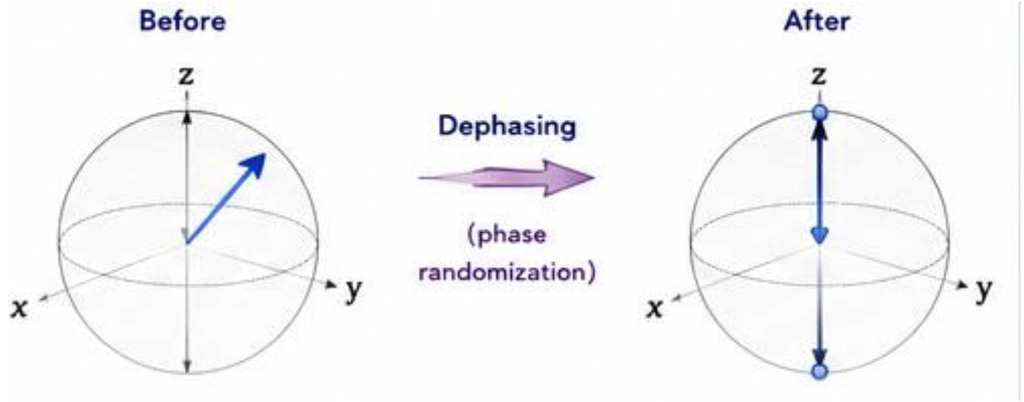




Outline: Talk #1

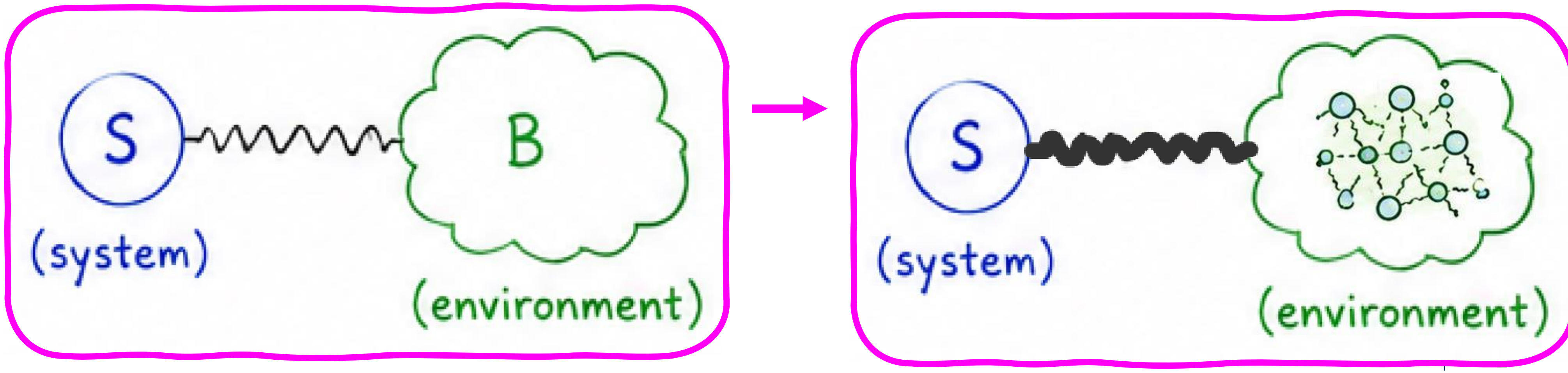
○ Intro

- Open Quantum Systems
- Density matrix, Interaction representation
- Spin Boson model
- T1 and T2 times



○ Derivation of the Quantum Master Equation:

- Redfield Equation
- Lindblad Equation
- Examples



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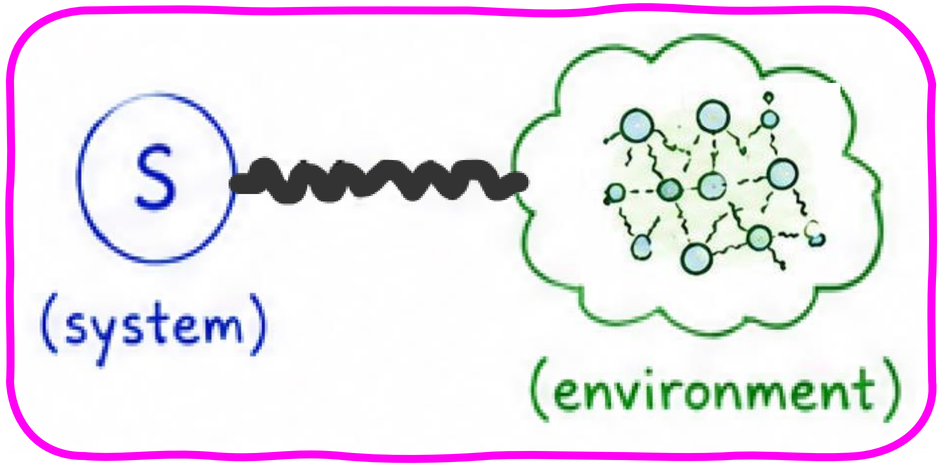
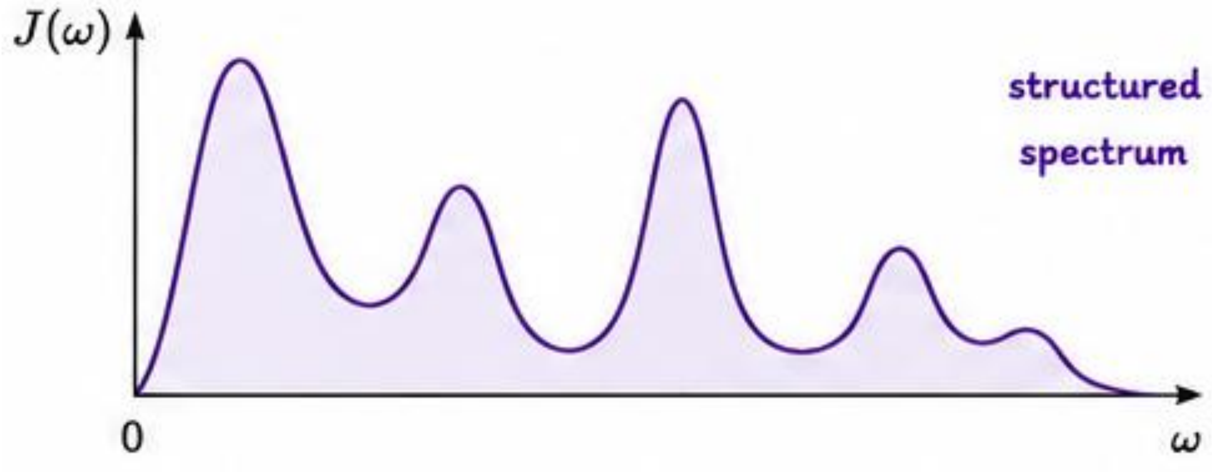
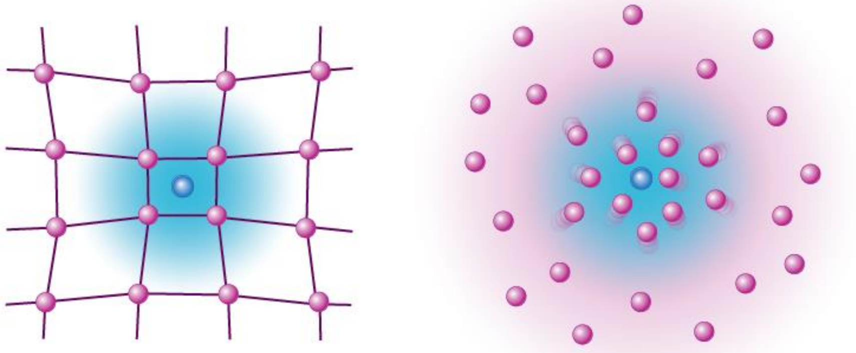
Outline: Talk #2



- **Review**
  - Weak coupling, Markovian approximation
  - Quantum master equation
- **Memory effects:** Measures, Methods
- **Strong coupling:** Polaron mapping
- **Other methods:** Repeated Interaction protocol, HEOM,…





F. Chevy, Physics 9, 86 (2016)

Time Evolution

1. **Undergraduate** in Chemistry and Computer Science
   (but I did not know I could combine them!)
   PhD in theoretical Chemistry

Tel Aviv University

2. **Postdoctoral fellowships**
   (quantum control, quantum dynamics and transport)

Weizmann Institute of Science

Columbia University

3. **Faculty** position in Chemistry (2007) and Physics (2019)

University of Toronto

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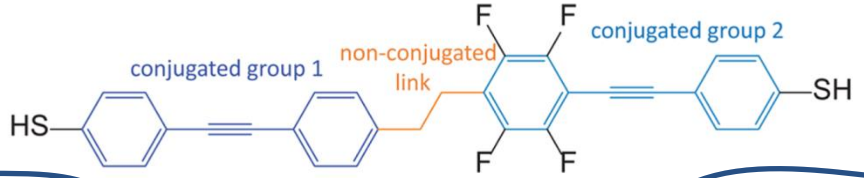
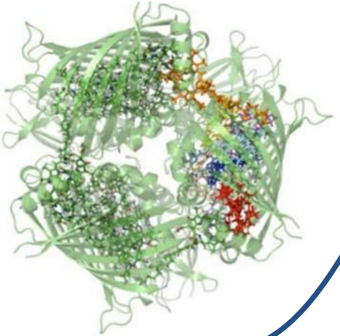
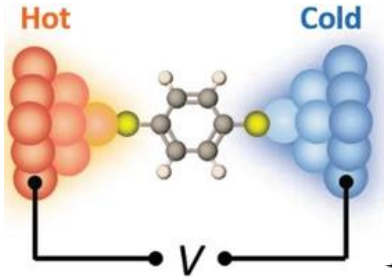
Research Evolution

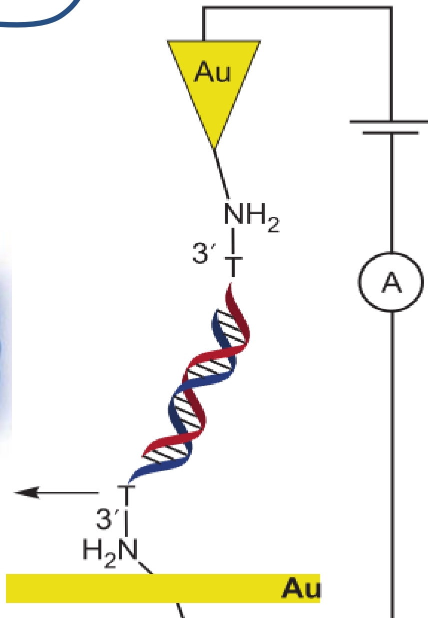
→ electron transfer reactions/processes

→ nanoelectronic devices





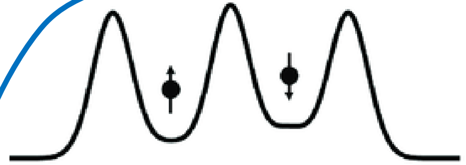




Nature Chemistry 7 221 (2015)

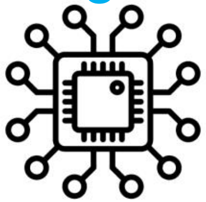
→ thermal energy management in electronic devices

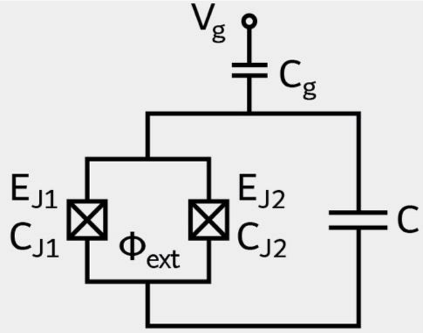
→ quantum charge and heat transport

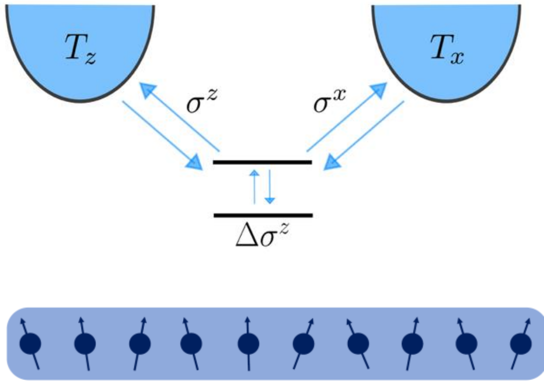


→ Minimal models -- ”Open quantum systems”

→ Applications to ”quantum technologies”







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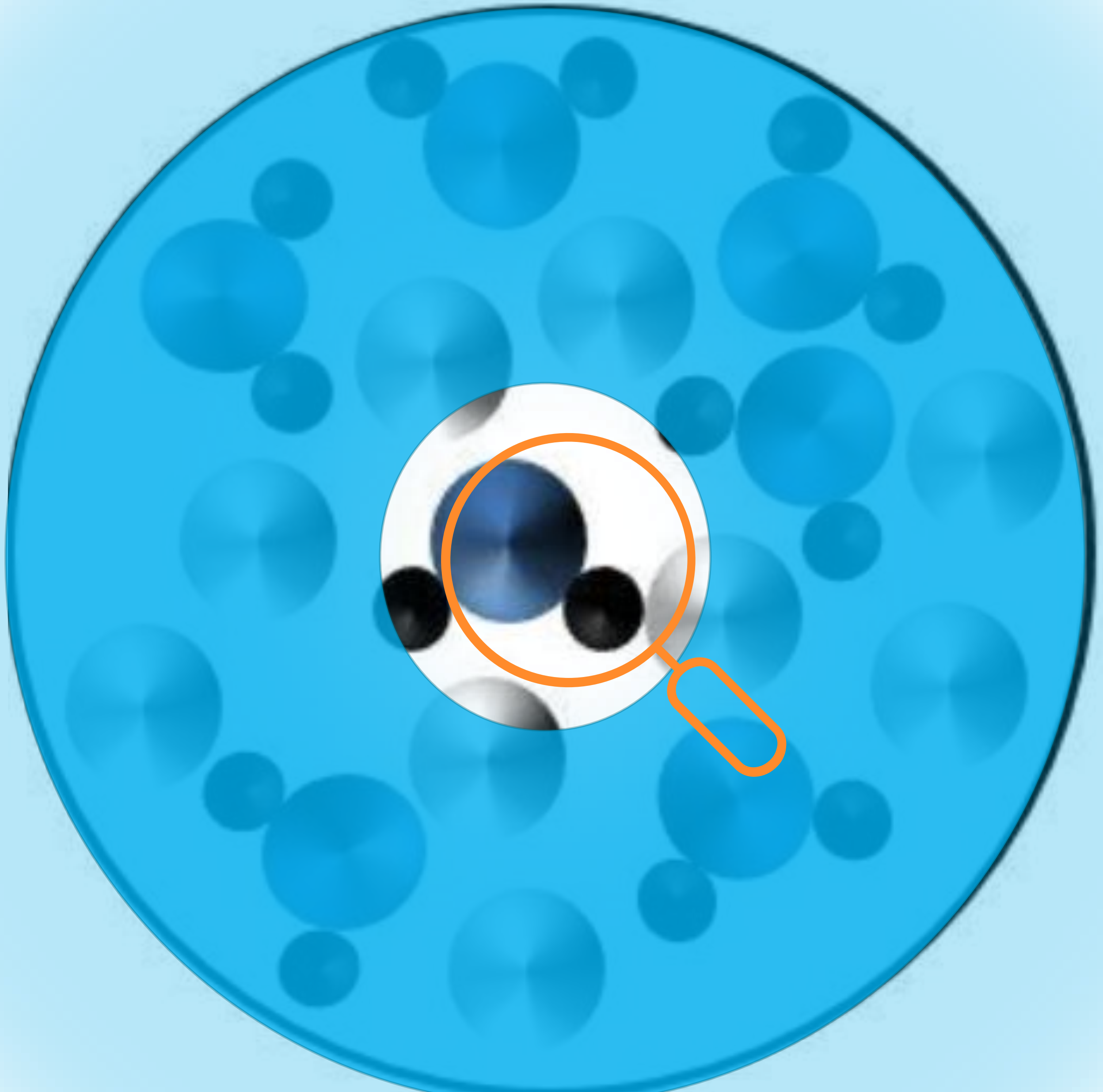
Theory of Open Quantum systems

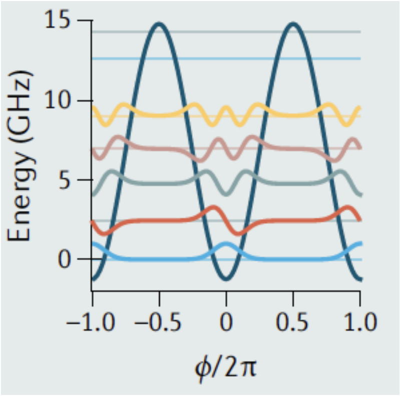
Fundamental and applications

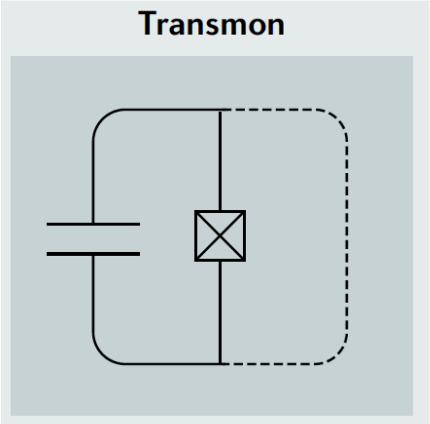
Method 1:
Solve accurately for all degrees of freedom, then focus.

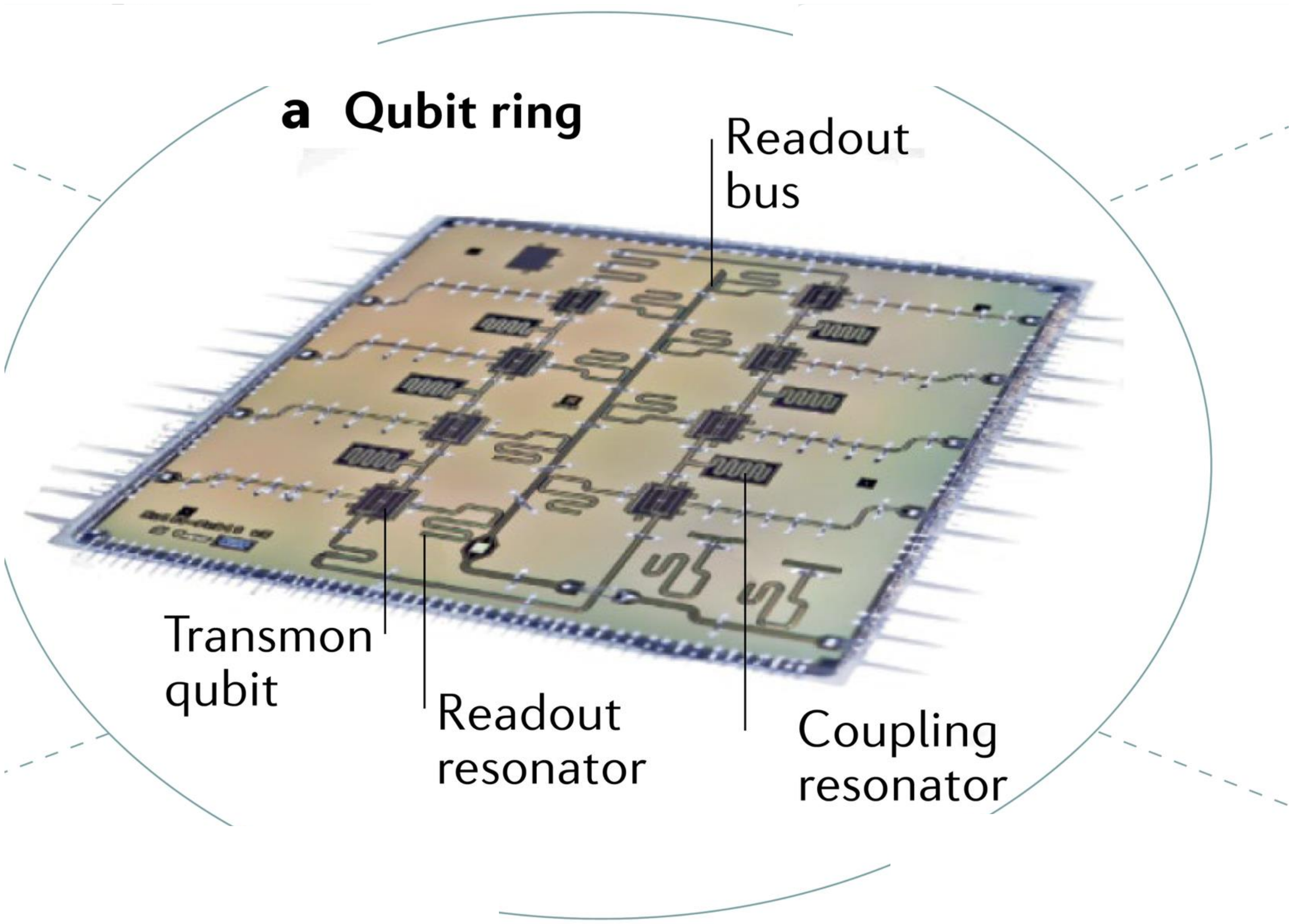
Method 2: Focus-solve accurately for the part that interests me while treating the rest approximately and with less details.



A superconducting qubit (the system) interacts with an environment composed of different physical sources

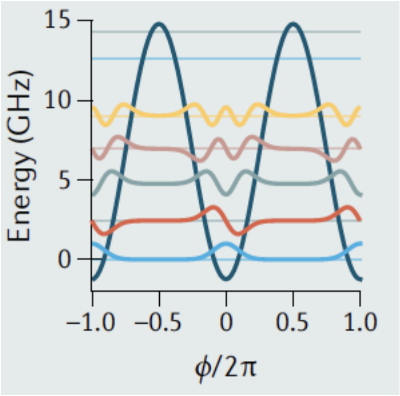


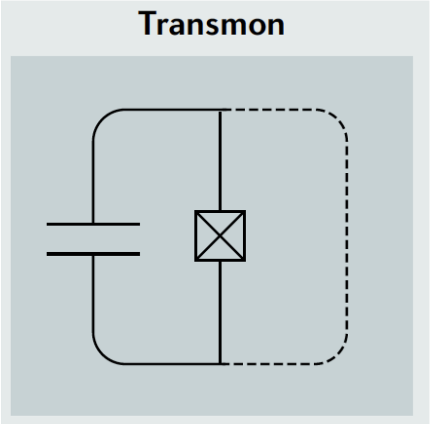


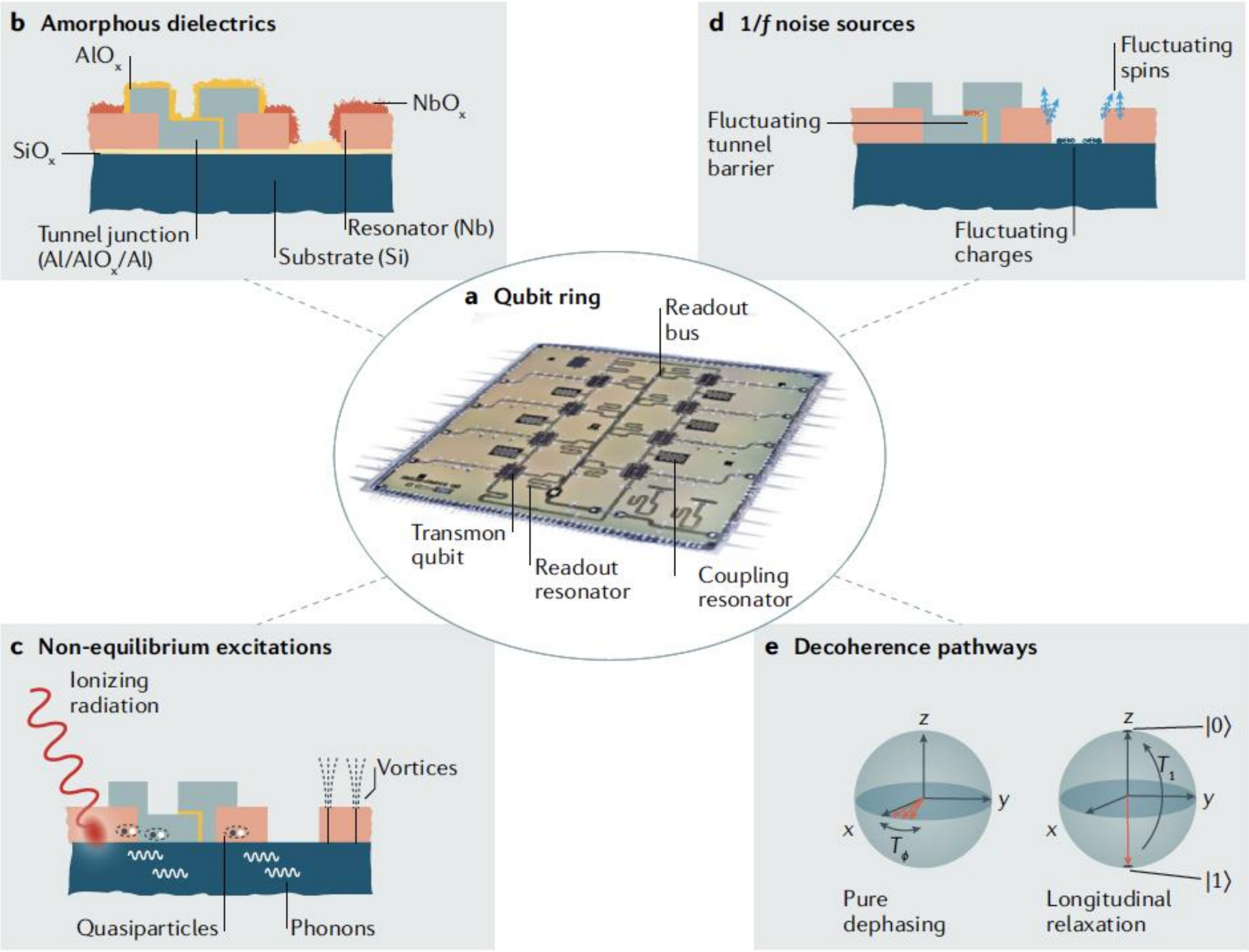


Ifran Siddiqi Nature Reviews Materials 6, 875 (2021)

A superconducting qubit (the system) interacts with an environment composed of different physical sources







a | Optical image of a qubit ring consisting of eight transmons. Each qubit is coupled to its neighbour via a resonator.

b surface and interfacial defect

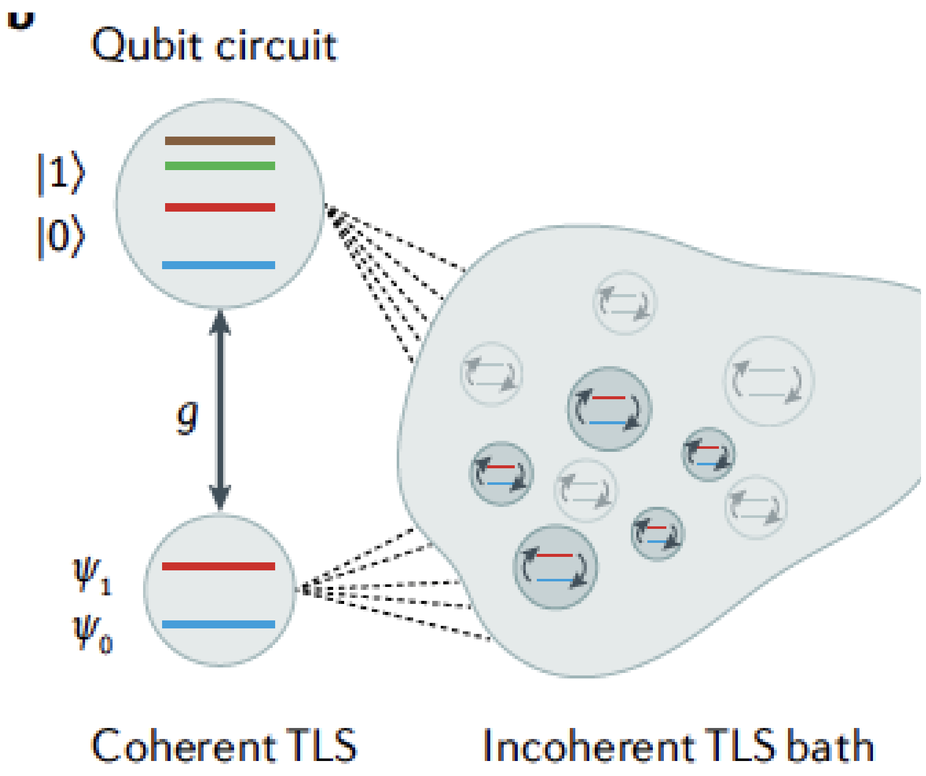
c | ionizing radiation and remnant magnetic fields

d | Microscopic charge and spin defects are believed to give rise to 1/f fluctuations

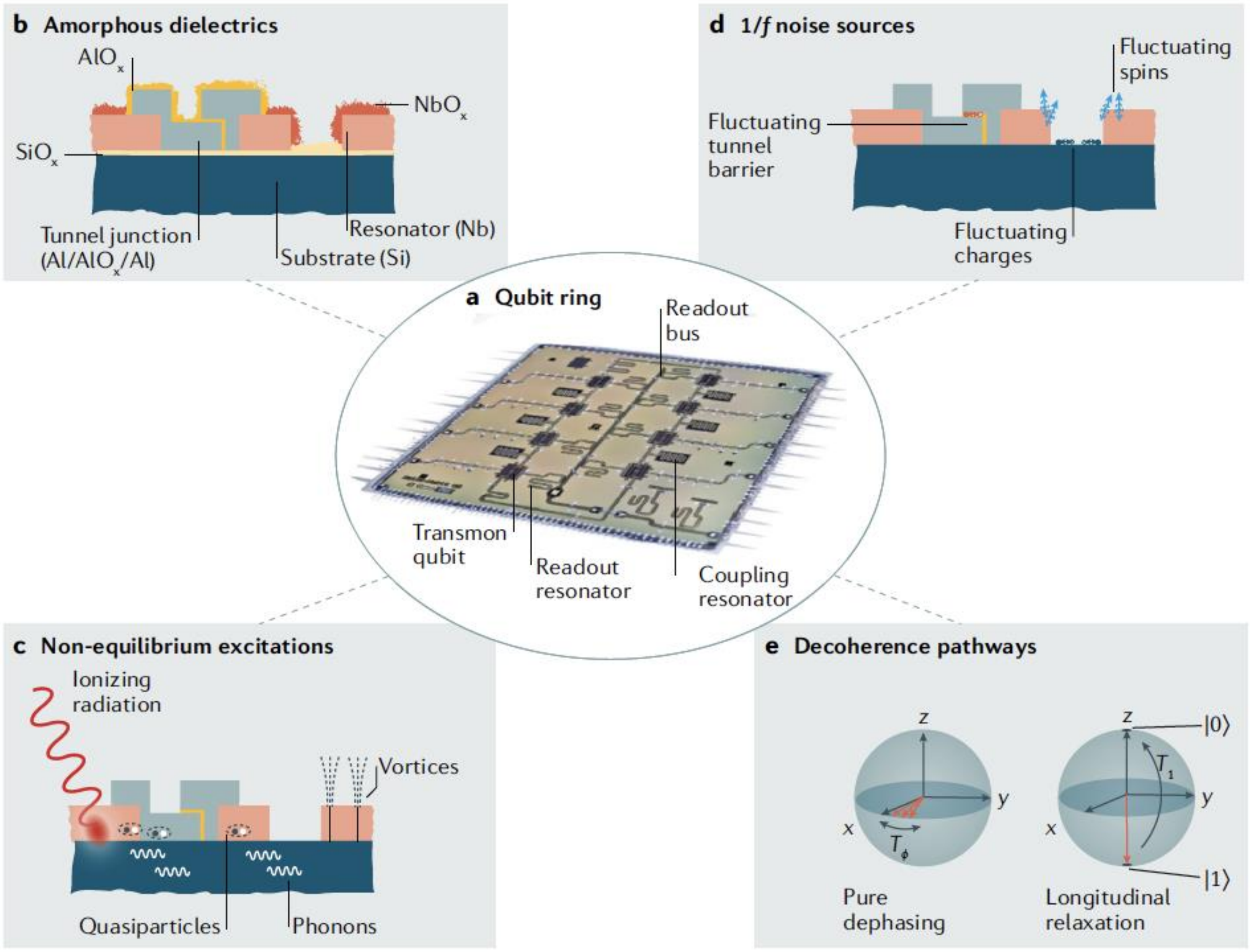
e | Fluctuations in the environment

Ifran Siddiqi Nature Reviews Materials 6, 875 (2021)

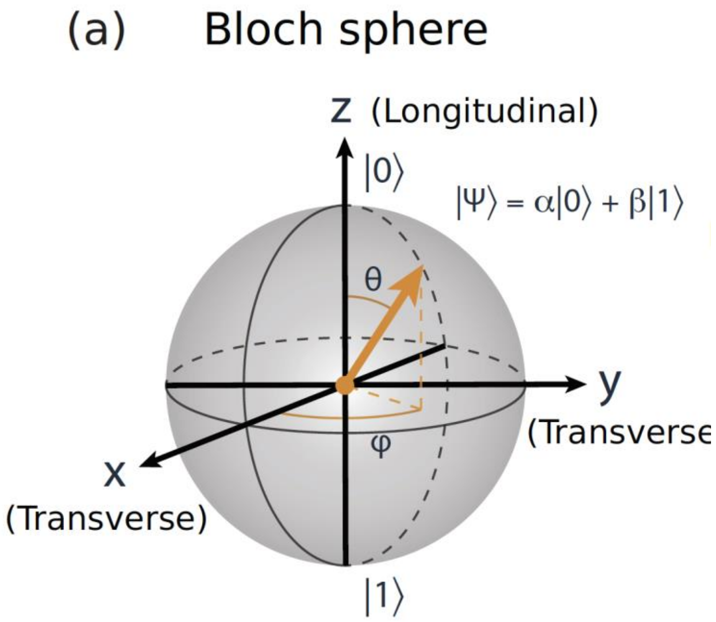
A superconducting qubit (the system) interacts with an environment composed of different physical sources



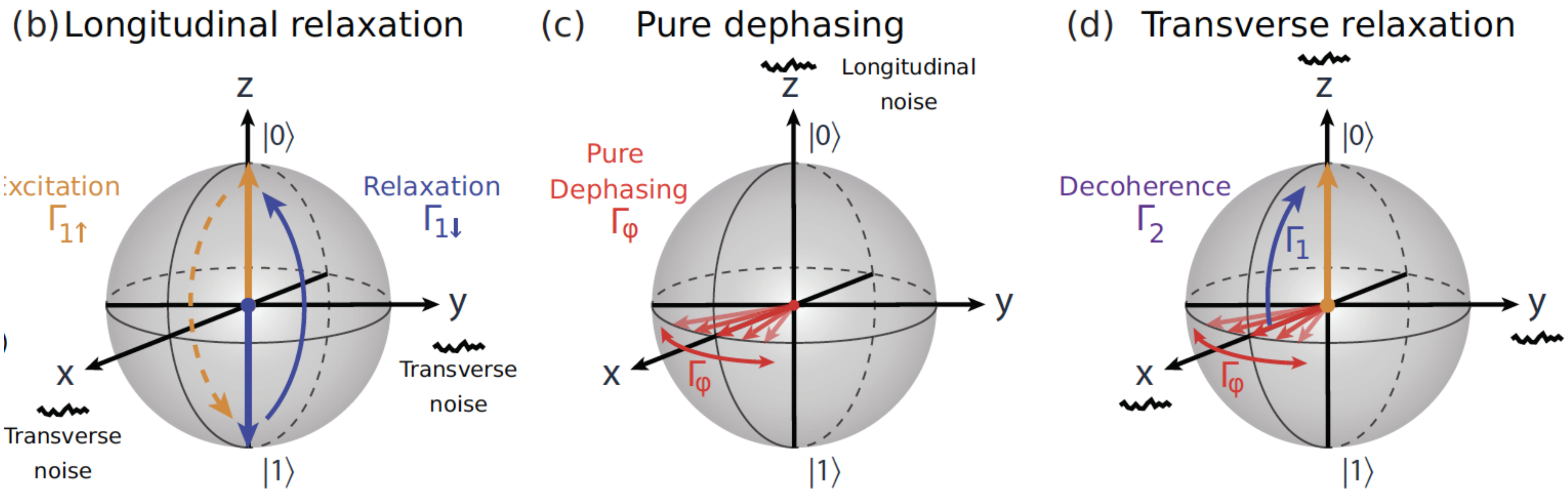
Ifran Siddiqi Nature Reviews Materials 6, 875 (2021)



Relaxation (T1) and Decoherence (T2)



$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$



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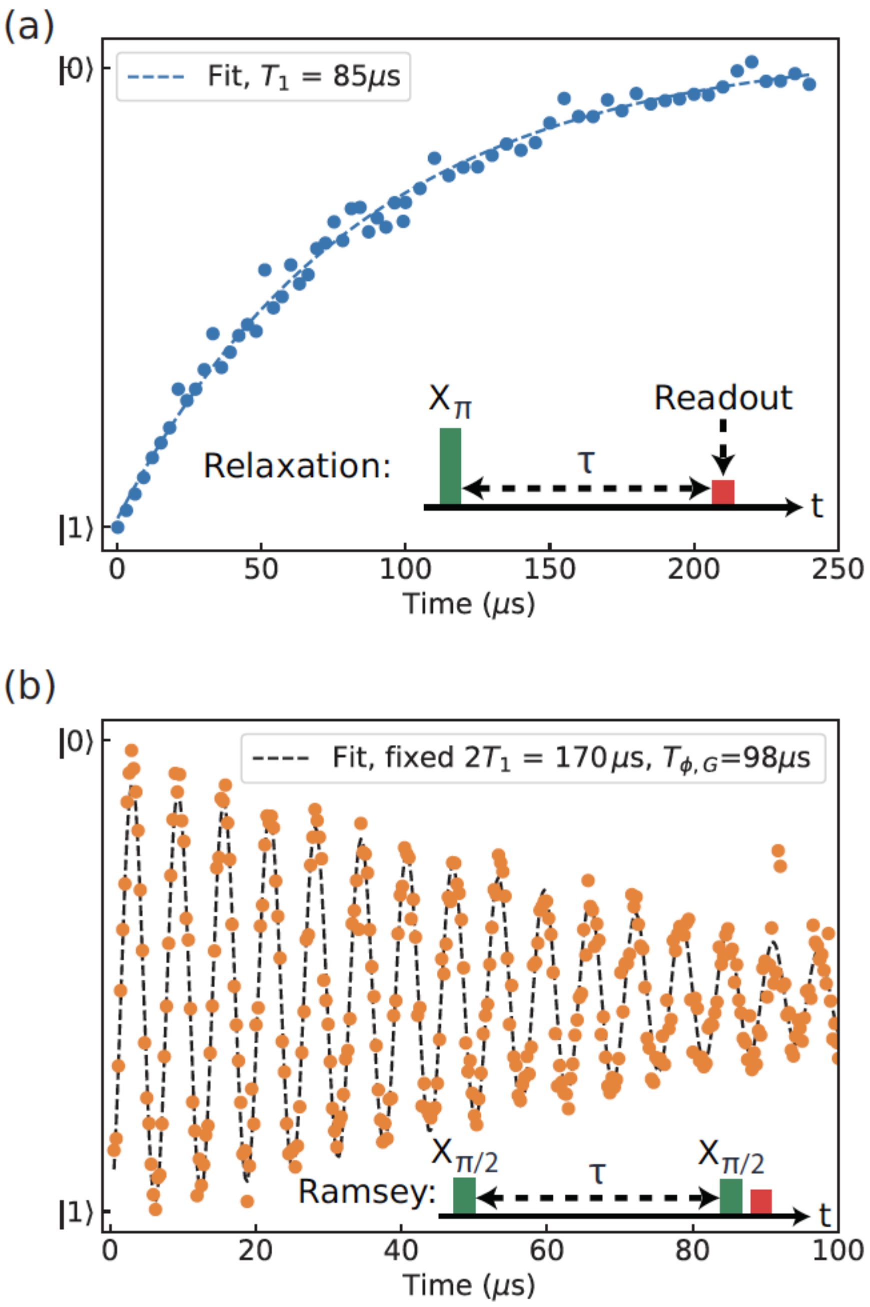
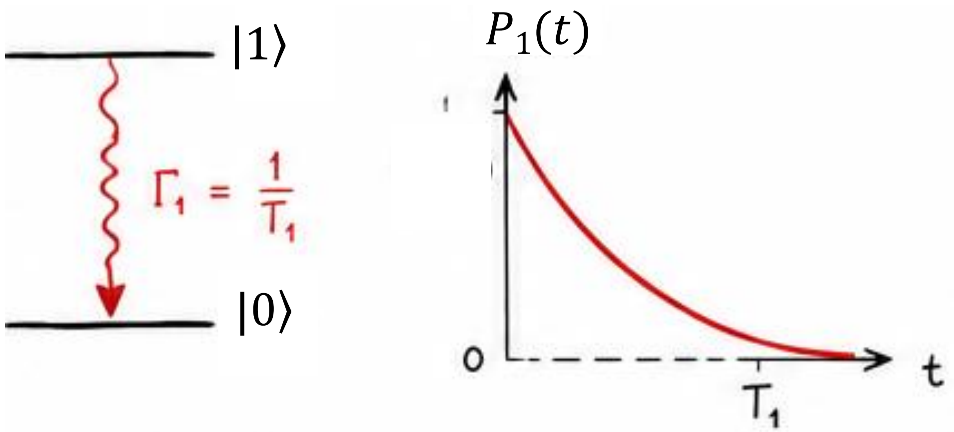
Relaxation (T1) and Decoherence (T2)



arXiv:1904.06560v5



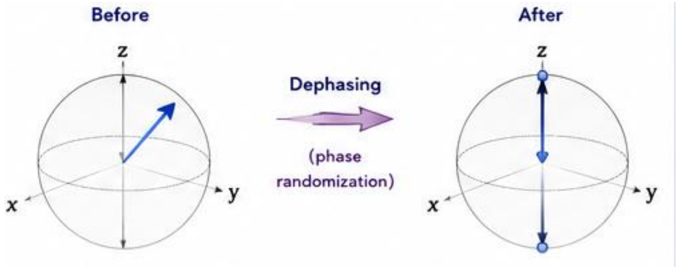
Benchmarking Quantum Gates and Circuits

Chem. Rev. 2025, 125, 5745–5775

Published as part of Chemical Reviews special issue "Quantum Computing".

Vinay Tripathi,[||] Daria Kowsari,[||] Kumar Saurav, Haimeng Zhang, Eli M. Levenson-Falk,[*]
and Daniel A. Lidar[*]

Single qubit gate errors



Coherent errors: Rotation error (the value of the angle is changed from the target), phase error (the axis of rotation is altered)

Incoherent errors:

- Relaxation (bit flip). typically modeled as **Markovian processes** leading to an exponential decay with characteristic time T1.
- Dephasing due to noise in the qubit frequency relative to a reference. These phase-flip errors may have **both Markovian and non-Markovian** components, and, in many condensed matter or solid-state systems, are typically modeled as a Markovian error plus a **classical noise with a 1/f power spectrum**

Leakage errors: the state evolving outside the qubit space (transmon supereconducting qubit)

Erasure: qubit information is lost (trapped atom qubits where an atom may escape the trap)

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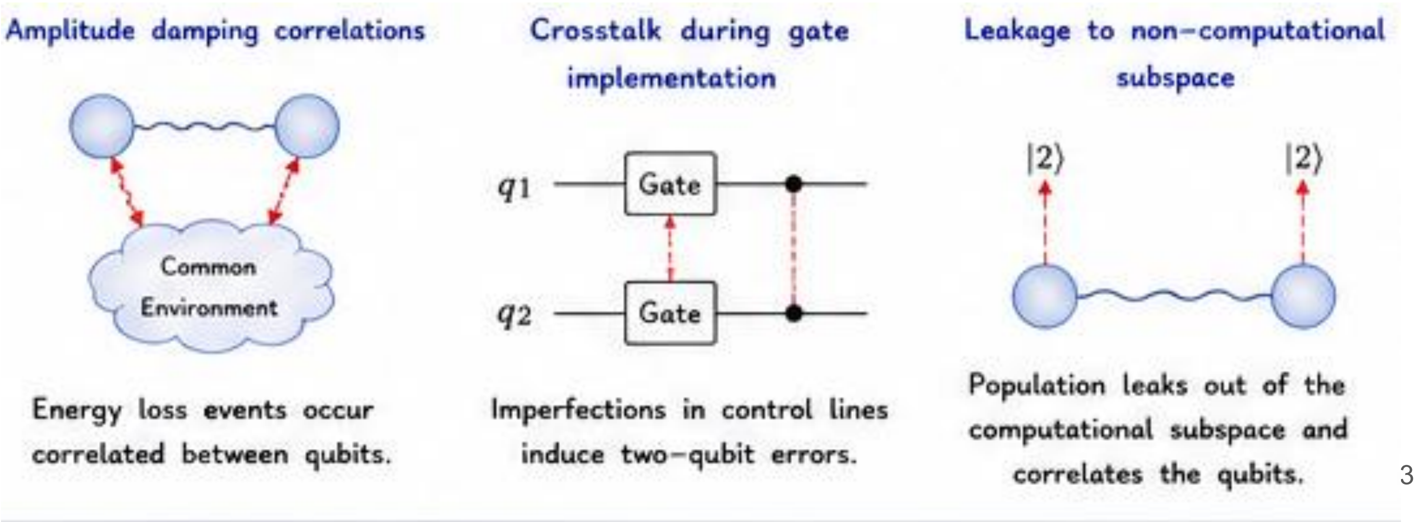
Vinay Tripathi,[‖] Daria Kowsari,[‖] Kumar Saurav, Haimeng Zhang, Eli M. Levenson-Falk,* and Daniel A. Lidar*

Two-qubit gate errors

iSWAP gate in transmon qubits:

Markovian relaxation and dephasing of each qubit; non-Markovian dephasing of each qubit, typically modeled as a 1/f qubit frequency noise spectrum; two qubit couplings to a Markovian or non-Markovian bath, e.g., a noisy tunable coupler; incorrect drive amplitude leading to transmon-transmon detuning and so causing incomplete swapping; two-qubit phase errors from unwanted ZZ interaction due to higher transmon levels; single-qubit phase errors due to improper calibration of level repulsion during transients at the beginning and end of the gate; and leakage into higher transmon levels or coupler states.

Examples of correlated errors:



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Relaxation (T1) and Decoherence (T2)

The density matrix of the qubit evolves according to a **quantum master equation of Lindblad form**

$$\dot{\rho}_S = -i\left[H_S, \rho_S\right] + \sum_{j=1}^{N} \Gamma_j \left(L_j \rho_S L_j^\dagger - \frac{1}{2}\left\{ L_j^\dagger L_j, \rho_S \right\} \right)$$

Coherent dynamics

Dissipative effects from the environment

Weak coupling
Markovian dynamics

Energy relaxation

$$L_1 = \sigma_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

$$\gamma_1 = 1/T_1$$

Causes decay $|1\rangle \rightarrow |0\rangle$ and contributes to both population and coherence decay.

Pure dephasing

$$L_\phi = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\gamma_\phi = 1/T_\phi$$

Causes random phase fluctuations (no energy exchange).

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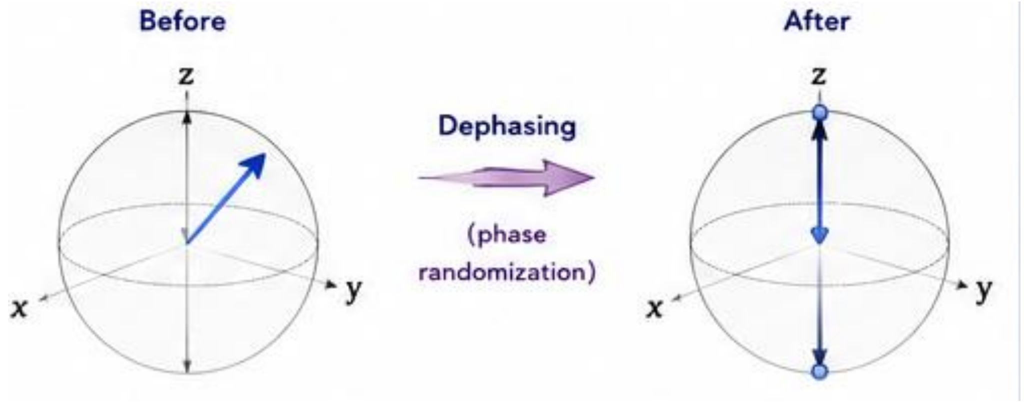
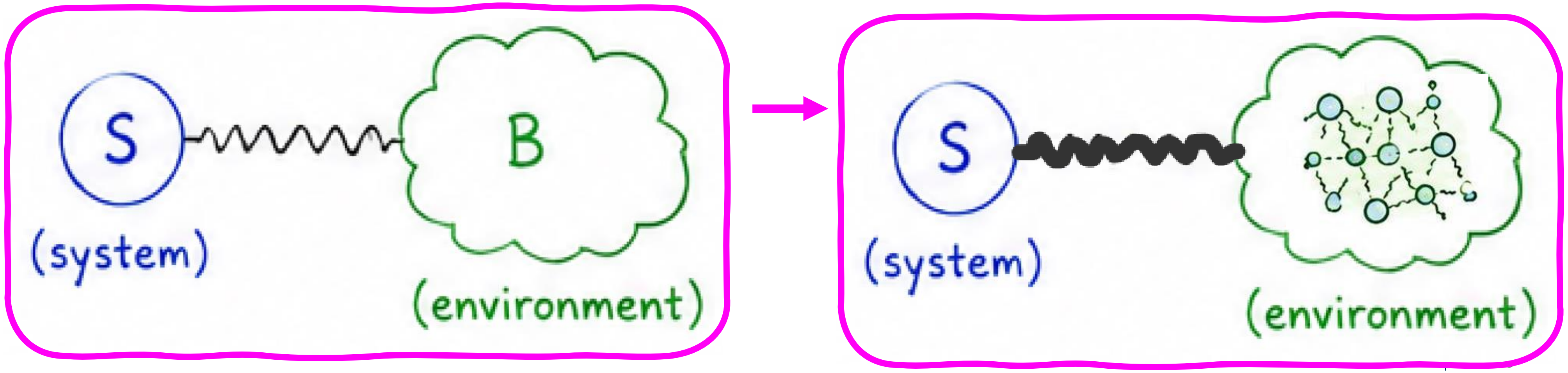
Outline: Talk #1

- **Intro**
  - Open Quantum Systems
  - Density matrix, Interaction Rep
  - Spin Boson model
  - T1 and T2 times



- **Derivation of the Quantum Master Equation:**
  - Redfield Equation
  - Lindblad Equation
  - Examples



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Background: Density matrix

Wavefunctions

A pure state may be expanded as

$$|\Psi(t)\rangle = \sum_n C_n(t)\,|n\rangle\,,$$

with normalization

$$\sum_n |C_n(t)|^2 = 1.$$

The expectation value of an observable A is

$$\langle A\rangle_t = \langle\Psi(t)|\,A\,|\Psi(t)\rangle\,.$$

Substituting the basis expansion yields

$$\langle A\rangle_t = \sum_{m,n} C_m^*(t)C_n(t)A_{mn},$$

where

$$A_{mn} = \langle m|\,A\,|n\rangle\,.$$

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Pure States

This motivates the definition of the density operator

$$\rho_S(t) = |\Psi(t)\rangle \langle\Psi(t)| \,.$$

Its matrix elements are

$$\rho_{mn}(t) = \langle m| \rho_S(t) |n\rangle = C_m(t)C_n^*(t).$$

The expectation value of any observable can then be written compactly as

$$\boxed{\langle A\rangle_t = \mathrm{Tr}\left[\rho_S(t)A\right] .}$$

The density operator satisfies

$$\mathrm{Tr}(\rho_S) = 1.$$

For a pure state,

$$\rho_S^2 = \rho_S.$$

This property is called *idempotency*.
Furthermore,

$$\rho_S^\dagger = \rho_S,$$

Density matrix – mixed state

The density operator becomes

$$\boxed{\rho_S = \sum_j P_j \left|\psi_j\right\rangle \left\langle\psi_j\right|.}$$

Statistical ensemble of pure states with statistical probabilities P_j

The probabilities satisfy

$$P_j \geq 0, \qquad \sum_j P_j = 1.$$

The expectation value of any observable remains

$$\langle A \rangle = \mathrm{Tr}\left(\rho_S A\right).$$

Quantum Liouville Equation

$$i\hbar \frac{\partial}{\partial t} \left|\Psi(t)\right\rangle = H_S \left|\Psi(t)\right\rangle \longrightarrow \boxed{\frac{d\rho_S}{dt} = -\frac{i}{\hbar}\left[H_S, \rho_S\right].}$$

$$\rho_S(t) = e^{-iH_S t/\hbar} \rho_S(0) e^{+iH_S t/\hbar}$$

Background: Interaction Representation

$$\hat{H} = \hat{H}_0 + \hat{V}$$

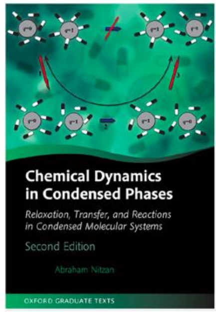
$$\langle \hat{O}(t) \rangle = \underbrace{\langle \psi | \hat{U}^\dagger(t)}_{\text{Schrödinger}} \; \hat{O} \; \underbrace{\hat{U}(t) |\psi\rangle}_{\text{Schrödinger}} = \langle \psi | \underbrace{\hat{U}^\dagger(t) \hat{O} \hat{U}(t)}_{\text{Heisenberg}} |\psi\rangle$$

$$\begin{aligned} \langle \hat{O}(t) \rangle &= \langle \psi | \hat{U}^\dagger(t) \mathbb{1} \hat{O} \mathbb{1} \hat{U}(t) |\psi\rangle \\ &= \underbrace{\langle \psi | \hat{U}^\dagger(t) e^{-i\hat{H}_0 t/\hbar}}_{\text{State}} \underbrace{e^{+i\hat{H}_0 t/\hbar} \hat{O} e^{-i\hat{H}_0 t/\hbar}}_{\text{Operator}} \underbrace{e^{+i\hat{H}_0 t/\hbar} \hat{U}(t) |\psi\rangle}_{\text{State}} \end{aligned}$$

$$\hat{\rho}_I(t) = e^{i\hat{H}_0 t/\hbar} \hat{\rho}_S(t) e^{-i\hat{H}_0 t/\hbar} = e^{i\hat{H}_0 t/\hbar} \underbrace{e^{i\hat{H} t/\hbar} \hat{\rho}(0) e^{-i\hat{H} t/\hbar}}_{\text{Schrödinger}} e^{-i\hat{H}_0 t/\hbar}$$

Background: Interaction Representation

Book of A. Nitzan



Schrödinger representation	Heisenberg representation	Interaction representation
$\Psi_S(t) = e^{-i\hat{H}t/\hbar}\Psi(t=0)$	$\Psi_H(t) = e^{i\hat{H}t/\hbar}\Psi_S(t) = \Psi_S(t=0)$	$\Psi_I(t) = e^{i\hat{H}_0 t/\hbar}\Psi_S(t)$
$\frac{\partial \Psi_S}{\partial t} = -(i/\hbar)\hat{H}\Psi_S$	(time independent)	$\frac{\partial \Psi_I}{\partial t} = -(i/\hbar)\hat{V}_I(t)\Psi_I$
$\hat{A}_S = \hat{A}(t=0)$	$\hat{A}_H(t) = e^{(i/\hbar)\hat{H}t}\hat{A}_S e^{-(i/\hbar)\hat{H}t}$	$\hat{A}_I(t) = e^{(i/\hbar)\hat{H}_0 t} A_S e^{-(i/\hbar)\hat{H}_0 t}$
(time independent)	$\frac{d\hat{A}_H(t)}{dt} = \frac{i}{\hbar}[\hat{H}, \hat{A}_H(t)]$	$\frac{d\hat{A}_I(t)}{dt} = \frac{i}{\hbar}[\hat{H}_0, \hat{A}_I(t)]$
$\hat{\rho}_S(t) = e^{-(i/\hbar)\hat{H}t}\hat{\rho}(0)e^{(i/\hbar)\hat{H}t}$	$\hat{\rho}_H = e^{(i/\hbar)\hat{H}t}\hat{\rho}_S(t)e^{-(i/\hbar)\hat{H}t}$	$\hat{\rho}_I(t) = e^{i\hat{H}_0 t/\hbar}\hat{\rho}_S(t)e^{-i\hat{H}_0 t/\hbar}$
$\frac{d}{dt}\hat{\rho}_S(t) = -\frac{i}{\hbar}[\hat{H}, \hat{\rho}_S(t)]$	$= \hat{\rho}(t=0)$ (time independent)	$\frac{d\hat{\rho}_I}{dt} = -\frac{i}{\hbar}[\hat{V}_I(t), \hat{\rho}_I(t)]$

From Chemical Physics to Quantum Technologies

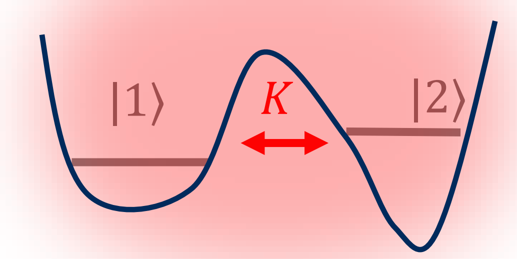
Variational calculation of the dynamics of a two level system interacting with a bath

J. Chem. Phys. 80, 2615 (1984)

Robert Silbey
Department of Chemistry, Massachusetts Institute of Technology, Cambridge, Massachusetts 02139
Robert A. Harris
Department of Chemistry, University of California, Berkeley, California 94720

(Received 18 November 1983; accepted 16 December 1983)

$$H = K\sigma_x + \sum_l \omega_l (a_l^+ a_l + 1/2) + \sigma_z \sum_l g_l (a_l + a_l^+)$$



Objective: rate of crossing between the two wells
= rate of reaction

What they did: Studied dynamics of a particle in a double well potential with the medium as an ideal gas of harmonic oscillators.

Context in Chemistry: Proton transfer reactions, energy transfer between chromophores in solution, reaction rate theory, solvation, electron transfer reactions, tunneling in hydrogen bonded structures.

Context in "Quantum": Qubit dynamics in an engineered environment, cavity or when coupled to nanoresonators, quantum energy transport

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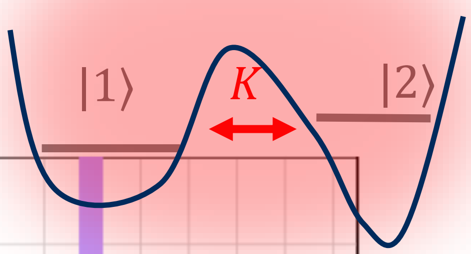
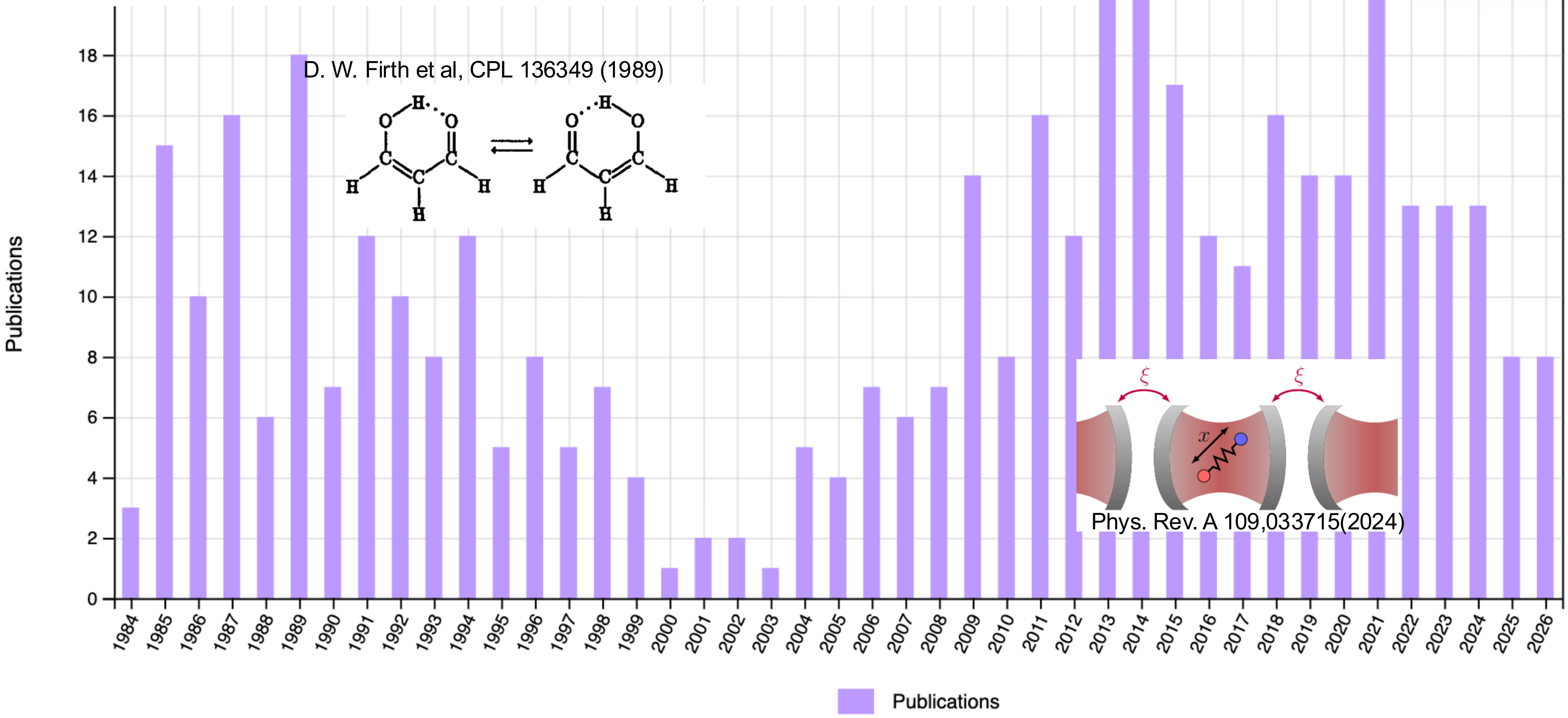
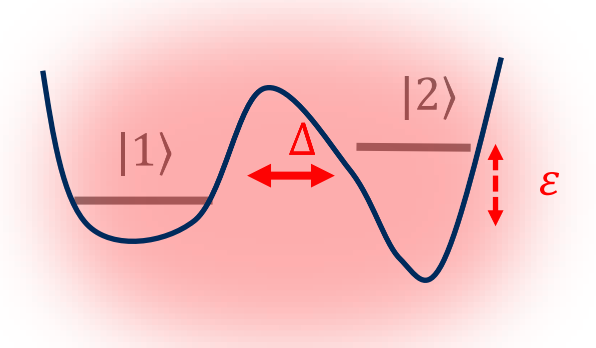




D. W. Firth et al, CPL 136349 (1989)

Phys. Rev. A 109,033715(2024)

The spin boson model



$$H_s = \epsilon_0 \sigma_z + \Delta \sigma_x$$

$$H_b = \sum_j \frac{1}{2}\left(p_j^2 + \omega_j^2 q_j^2\right) \longrightarrow \rho_b = \frac{e^{-\beta H_b}}{\mathrm{Tr}_b\left\{e^{-\beta H_b}\right\}}$$

$$\boxed{\begin{aligned}\sigma_x &= |1\rangle\langle 2| + |2\rangle\langle 1| \\ \sigma_z &= |1\rangle\langle 1| - |2\rangle\langle 2|\end{aligned}}$$

$$H_{sb} = \sigma_z \sum_j g_j q_j$$

$$J(\omega) = \frac{\pi}{2}\sum_j \frac{g_j^2}{\omega_j}\delta\left(\omega - \omega_j\right) \longrightarrow J(\omega) = \frac{E_r}{2}\frac{\omega\omega_c}{\omega^2 + \omega_c^2}$$

The Spin boson model

Model:

$$H_s = \epsilon_0 \sigma_z + \Delta \sigma_x$$

$$H_b = \sum_j \frac{1}{2}\left(p_j^2 + \omega_j^2 q_j^2\right)$$

$$H_{sb} = \sigma_z \sum_j g_j q_j$$

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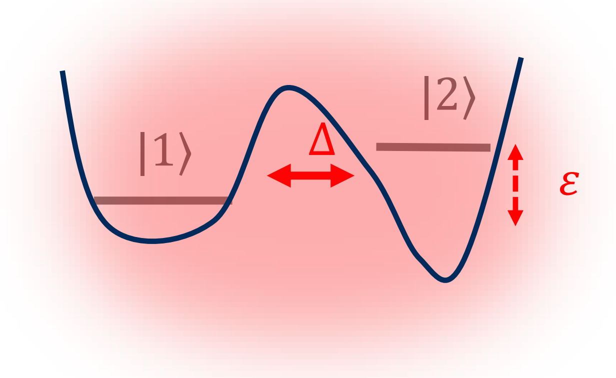


Dynamics:

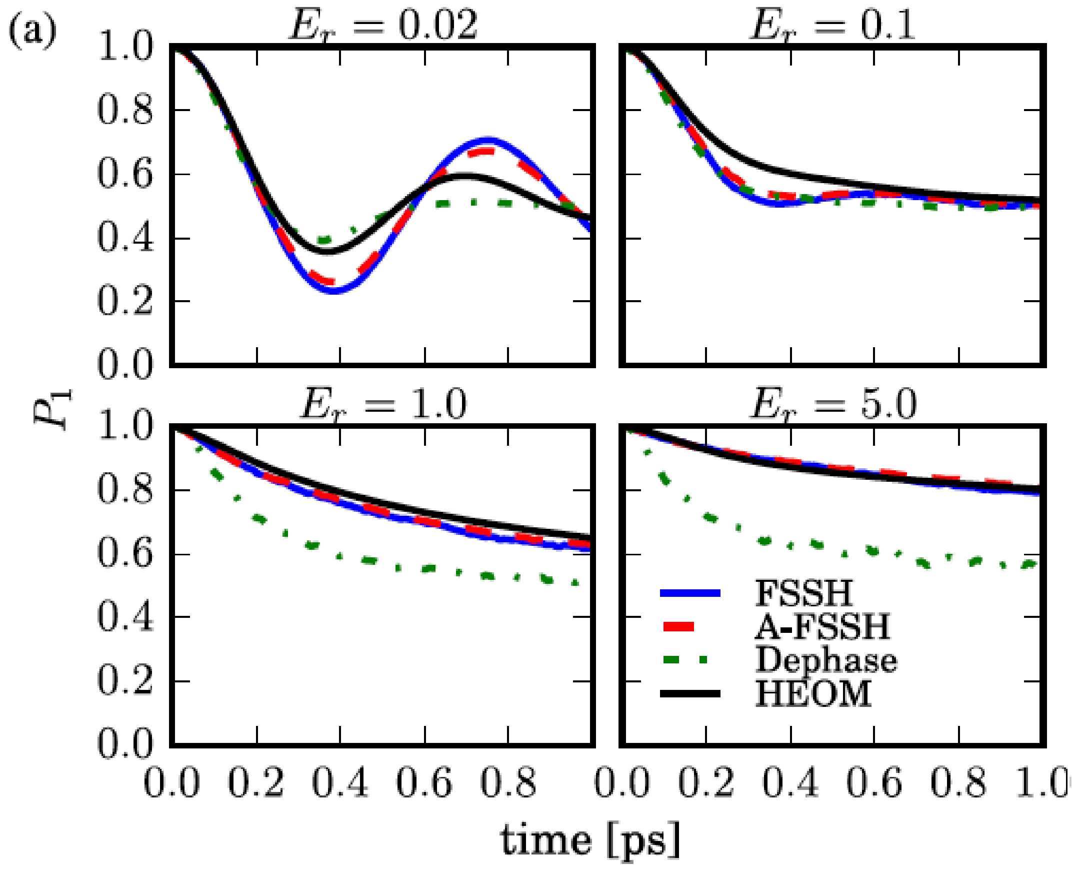
Initial condition $\rho_b \left|1\right\rangle \left\langle 1\right|$

$$\langle \sigma_z(t) \rangle = \mathrm{Tr}\left\{ \rho_0 e^{iHt} \hat{\sigma}_z e^{-iHt} \right\}$$

THE JOURNAL OF CHEMICAL PHYSICS **144**, 094104 (2016)

On the accuracy of surface hopping dynamics in condensed phase non-adiabatic problems

Hsing-Ta Chen and David R. Reichman
Department of Chemistry, Columbia University, New York, New York 10027, USA

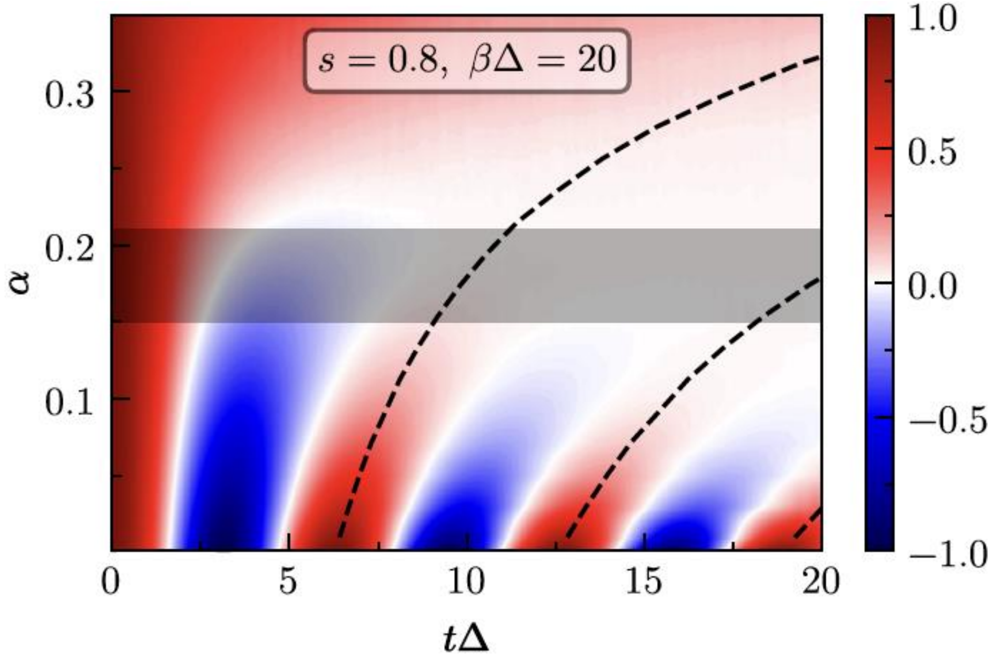


PHYSICAL REVIEW LETTERS **134**, 056502 (2025)

Transient Dynamical Phase Diagram of the Spin-Boson Model

Olga Goulko,[1,*] Hsing-Ta Chen,[2,†] Moshe Goldstein,[3,‡] and Guy Cohen[4,§]

$$\langle \sigma_z(t) \rangle = \mathrm{Tr}\left\{ \rho_0 e^{iHt} \hat{\sigma}_z e^{-iHt} \right\}$$



QMC simulations.
Time evolution of polarization as a function of coupling α close to the Ohmic regime. The horizontal shaded regions indicate the approximate location of the localization-delocalization transition α extracted from dynamical data.

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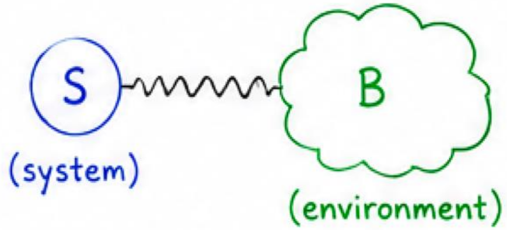
Theory of Open Quantum systems

Fundamental and applications

1. SETUP

A quantum system S interacts with an environment (bath) B.



Total Hamiltonian

$$H = H_S + H_B + H_{SB}$$

2. Questions

1. What are the roles of quantum effects in “condensed phases”?
2. Can dissipation generate useful quantum phenomena (dissipation state engineering)
3. What are the tradeoffs and limits in quantum thermodynamics and quantum computing
4. How do many body open quantum systems behave?
5. How does classical behaviour emerges from quantum mechanics?

- Impact of noise of quantum dynamics
- Strong coupling
- Nonmarkovianty (structured bath)
- Electron-phonons-spins interactions

4. Toolbox

Quantum mechanics
Many body physics
Computational Physics
(quantum) statistical mechanics
(quantum) Thermodynamics

3. Applications:

Reaction dynamics – chemical Physics
Condensed matter Physics
Quantum optics
Quantum thermodynamics
Quantum information science –quantum technologies

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Theory of Open Quantum systems

Fundamental and applications

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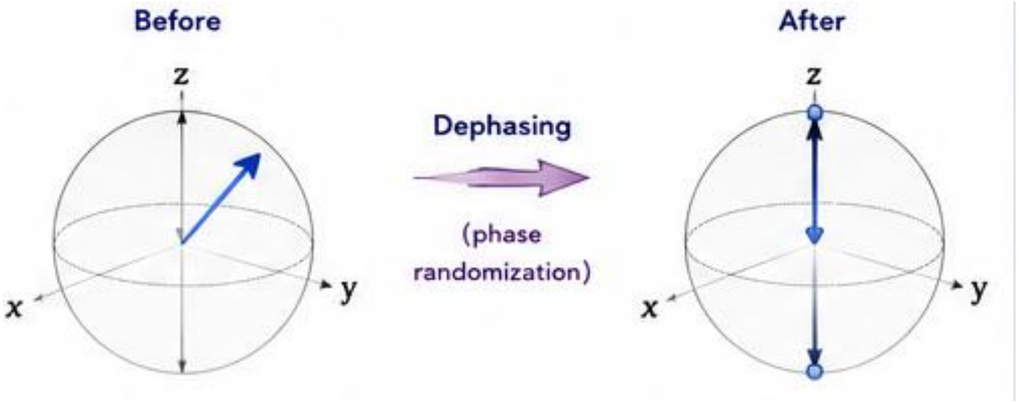
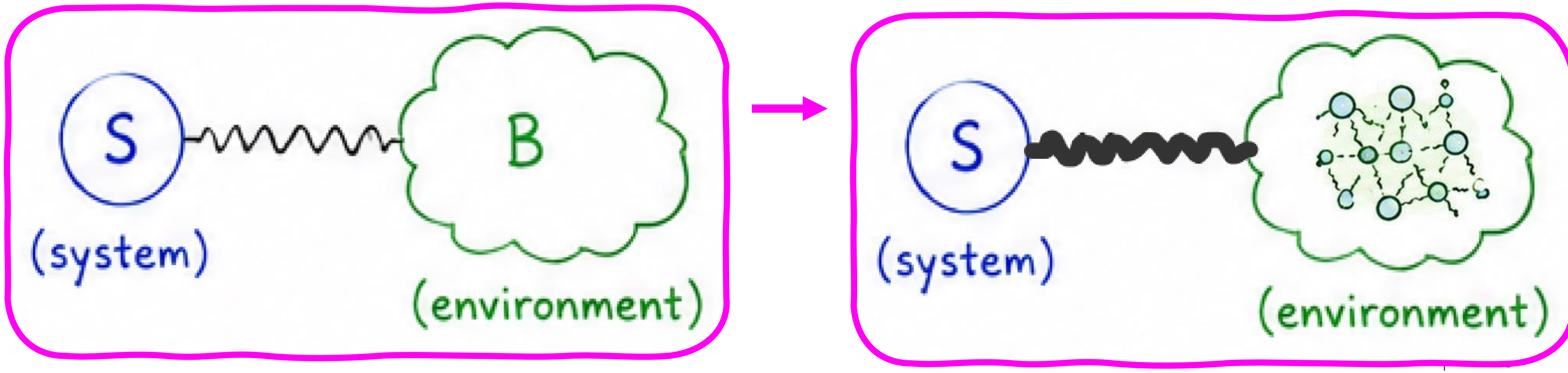
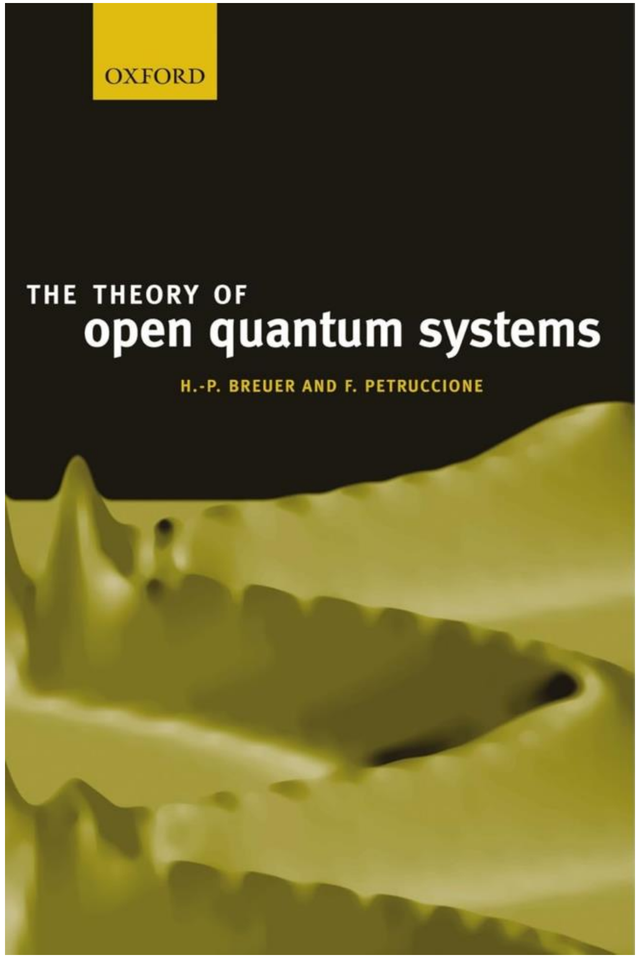
Outline: Talk #1

- **Intro**
  - Open Quantum Systems
  - Density matrix, Interaction Rep
  - Spin Boson model
  - T1 and T2 times



- **Derivation of the Quantum Master Equation:**
  - Redfield Equation
  - Lindblad Equation
  - Examples





II DENSITY MATRIX THEORY

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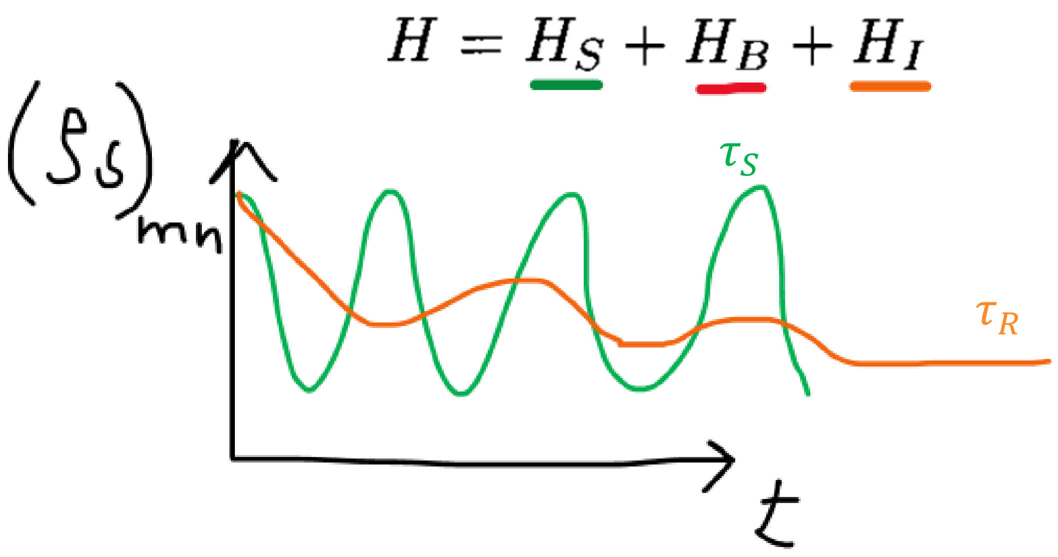
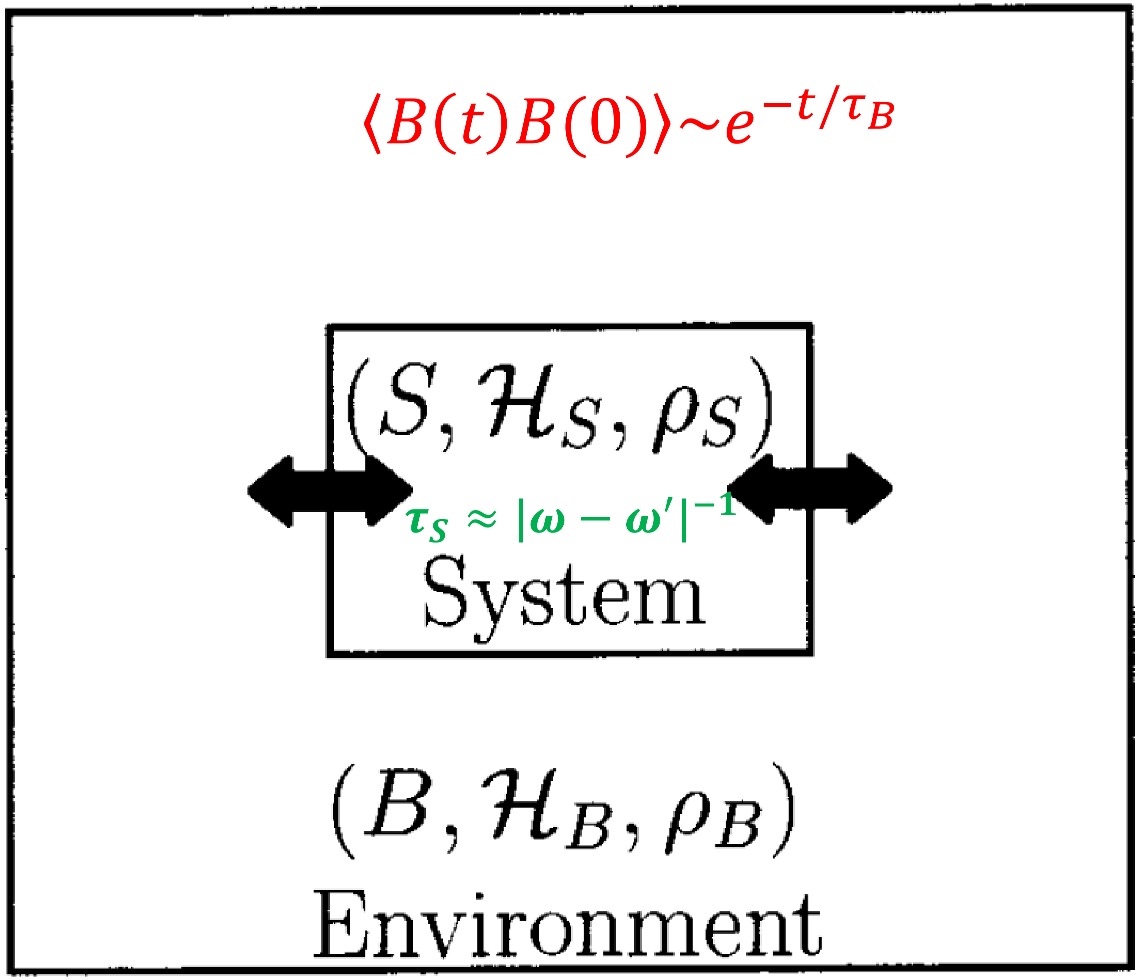
Weak coupling, Markovian, population --- coherence

Quantum master equations: Microscopic derivation





$\tau_S, \tau_R \gg \tau_B$ The bath dynamics is fast – faster than any other timescales of the system

$\tau_R \gg \tau_S$ The relaxation dynamics is slow compared to the (pure) dynamics of the system

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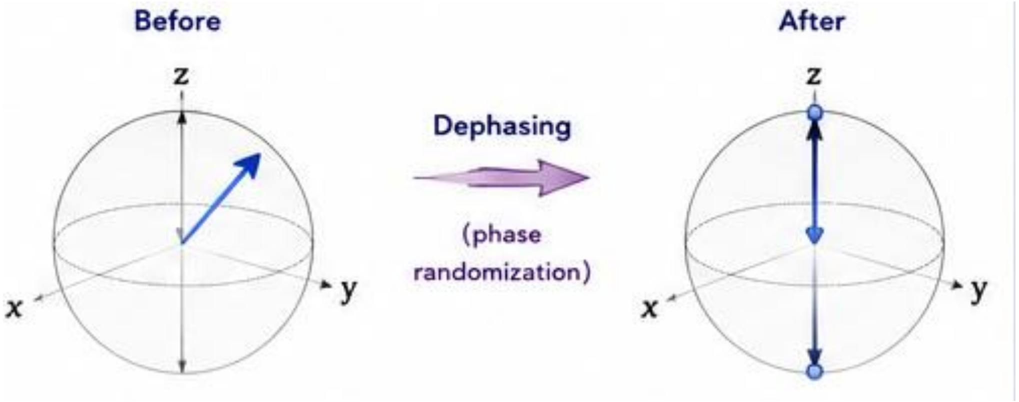
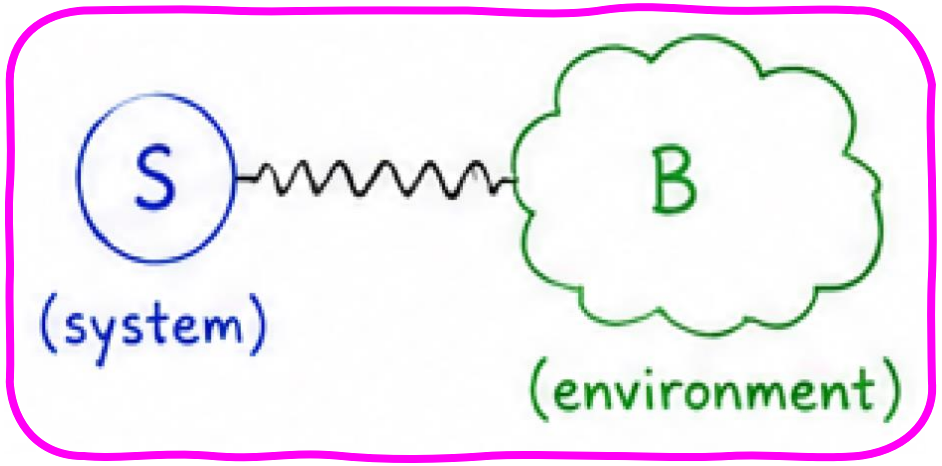
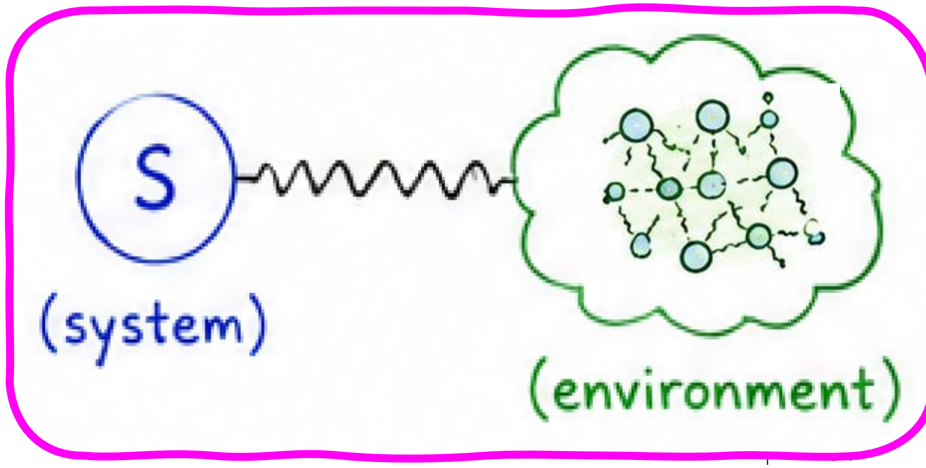


- **Derivation of the Quantum Master Equation:**
  - Redfield Equation
  - Lindblad Equation
  - Examples





Lindblad Equation: Properties

$$\rho_S(t) = \mathrm{Tr}_B\left[\rho_{\mathrm{tot}}(t)\right]$$

The general Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) equation is

$$\boxed{\frac{d\rho_S}{dt} = -\frac{i}{\hbar}[H_S, \rho_S] + \sum_j \gamma_j \left(L_j \rho_S L_j^\dagger - \frac{1}{2}\left\{ L_j^\dagger L_j, \rho_S \right\} \right)}$$

- ρ_S is the reduced density matrix,
- H_S is the system Hamiltonian,
- L_j are Lindblad (jump) operators,
- γ_j are dissipation rates.

Lindblad Equation: Properties

$$\rho_S(t) = \mathrm{Tr}_B\left[\rho_{\mathrm{tot}}(t)\right]$$

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Coherent evolution

Quantum jump

Normalization

spontaneous emission,
thermal excitation,
leakage

University of Toronto

Lindblad Equation: Properties

The general Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) equation is

$$\boxed{\frac{d\rho_S}{dt} = -\frac{i}{\hbar}[H_S, \rho_S] + \sum_j \gamma_j \left(L_j \rho_S L_j^\dagger - \frac{1}{2} \left\{ L_j^\dagger L_j, \rho_S \right\} \right)}$$

Trace Preservation $\mathrm{Tr}(\rho_S) = 1$ Complete Positivity

Hermiticity Preservation $\rho_S^\dagger = \rho_S \longrightarrow \rho_S(t)^\dagger = \rho_S(t)$

Observable expectation values therefore remain real

Positivity $\langle \psi | \rho_S | \psi \rangle \geq 0$ for all states $|\psi\rangle$ **Complete Positivity**

Markovianity

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5 Example 1: Pure Dephasing

Consider a qubit described by

$$H_S = \frac{\omega_q}{2}\sigma_z.$$

Pure dephasing arises from fluctuations in the qubit energy splitting. The Lindblad operator is

$$\boxed{L = \sigma_z.}$$

The master equation becomes

$$\frac{d\rho_S}{dt} = -\frac{i}{\hbar}[H_S, \rho_S] + \gamma_\phi \left(\sigma_z \rho_S \sigma_z - \rho_S\right).$$

$$\rho_S = \begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix} \rightarrow \quad \dot{\rho}_{00} = 0, \qquad \dot{\rho}_{11} = 0$$

$$\searrow \quad \dot{\rho}_{01} = -2\gamma_\phi \rho_{01} \rightarrow \boxed{\rho_{01}(t) = \rho_{01}(0)e^{-2\gamma_\phi t}}$$

6 Example 2: Population Relaxation

Consider spontaneous emission

$$|1\rangle \rightarrow |0\rangle.$$

The Lindblad operator is

$$\boxed{L = \sigma_-.}$$

The master equation becomes

$$\frac{d\rho_S}{dt} = -\frac{i}{\hbar}[H_S, \rho_S] + \gamma \left(\sigma_- \rho_S \sigma_+ - \frac{1}{2}\{\sigma_+\sigma_-, \rho_S\} \right).$$

$$\dot{\rho}_{11} = -\gamma \rho_{11} \rightarrow \rho_{11}(t) = \rho_{11}(0) e^{-\gamma t} \rightarrow \boxed{T_1 = \frac{1}{\gamma}}$$

$$\rho_{01}(t) = \rho_{01}(0) e^{-\gamma t/2}$$

7 Example 3: Particle Loss

Consider a cavity mode

$$H_S = \omega_c a^\dagger a.$$

Photons leak from the cavity.
The Lindblad operator is

$$\boxed{L = a.}$$

The master equation becomes

$$\frac{d\rho_S}{dt} = -\frac{i}{\hbar}[H_S, \rho_S] + \kappa\left(a\rho_S a^\dagger - \frac{1}{2}\{a^\dagger a, \rho_S\}\right).$$

Mean Photon Number $n = \langle a^\dagger a\rangle$

$$\frac{dn}{dt} = -\kappa n \rightarrow \boxed{n(t) = n(0)e^{-\kappa t}}$$

8 Relaxation and Dephasing Together

For superconducting qubits both processes are typically present.

The master equation becomes

$$\frac{d\rho_S}{dt} = -\frac{i}{\hbar}[H_S, \rho_S] + \frac{1}{T_1}\mathcal{D}[\sigma_-] + \frac{1}{2T_\phi}\mathcal{D}[\sigma_z],$$

where

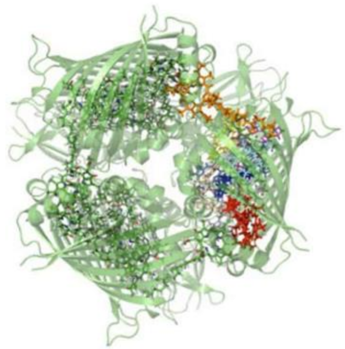
$$\mathcal{D}[L] = L\rho_S L^\dagger - \frac{1}{2}\left\{L^\dagger L, \rho_S\right\}.$$

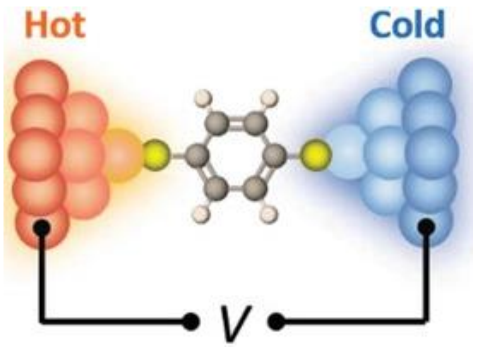
The coherence time satisfies

$$\boxed{\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\phi}}$$

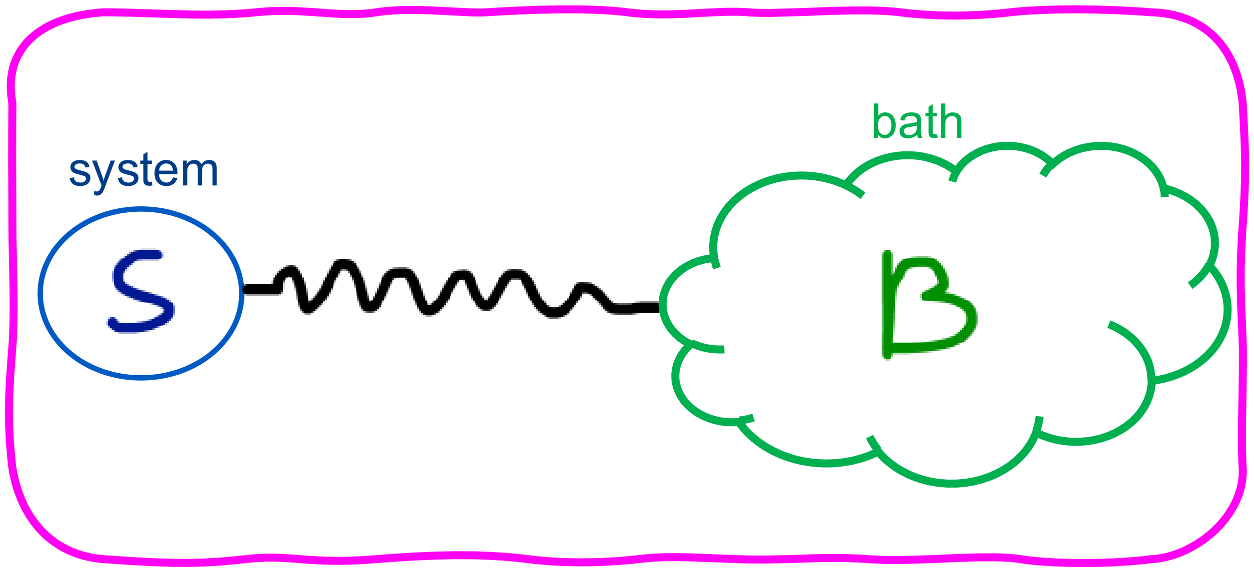
which is one of the most important relations in quantum information processing.

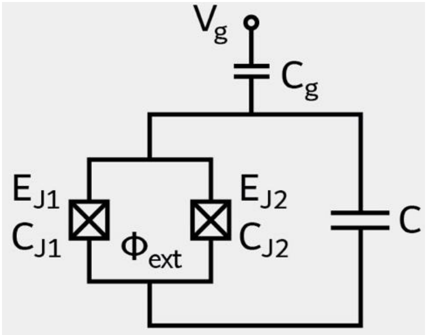
Open quantum systems: Summary

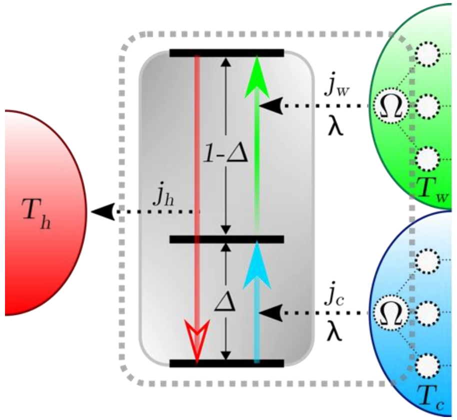


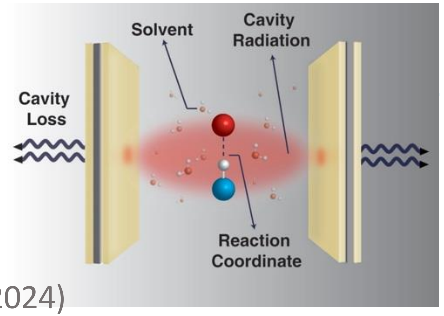


Advanced Functional Materials
Thermal and Thermoelectric Properties of Molecular Junctions
K Wang, E Meyhofer, P Reddy
https://doi.org/10.1002/adfm.201904534

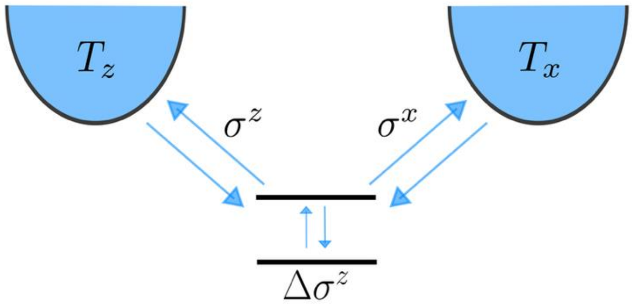


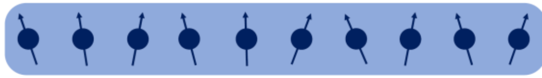






W Ying, P Huo
Communications Materials 5, 110 (2024)





Summary #1: Open quantum systems

Challenge to **go beyond "standard" approximations**:

Fundamentals:

Understand the role of quantum effects in "condensed phases"

- Biological (light harvesting complexes)
- chemical (chemical processes/reactions in solution, on surfaces)
- Processes in organic and inorganic materials (organic solar cells)
- engineered (quantum computing, quantum sensing)

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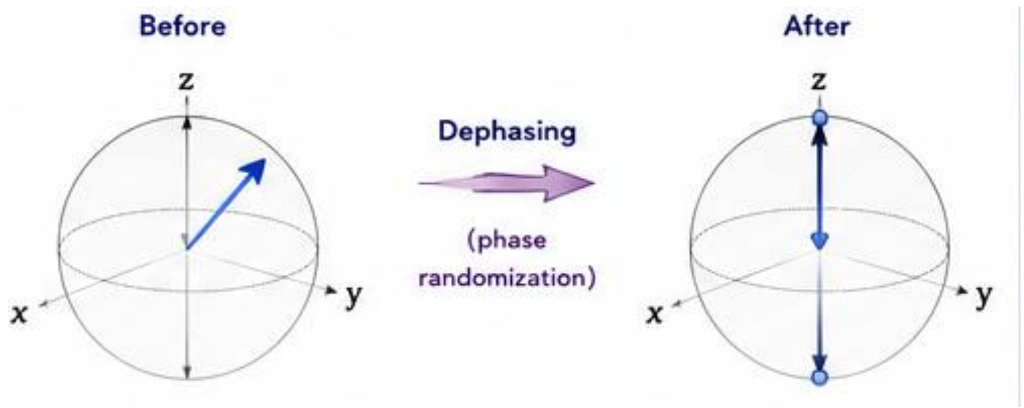

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